

PRIVATE KEY d -> HASH160 -> ADDRESS (solved puzzles)

Pipeline:

1.  $P = d * G$  (secp256k1 scalar multiply)
2. compressed pubkey = (02 if y even else 03) || x (33 bytes)
3. sha256 = SHA256(compressed pubkey)
4. hash160 = RIPEMD160(sha256) [NOT RIPEMD160(d)]
5. payload = 0x00 || hash160 || checksum4
6. address = Base58Check(payload)

solved keys: 82

=== summary table ===

|      | P   | bits | d_t3 | y%2 | h160_4 | h160_t4 | addr_head4 | addr                                |
|------|-----|------|------|-----|--------|---------|------------|-------------------------------------|
| P 1  | 1   | 1    | 1    | e   | 751e   | 3bd6    | 1BgG       | 1BgGZ9tcN4rm9KBzDn7KprQz87SZ26SAMH  |
| P 2  | 2   | 2    | 3    | e   | 7dd6   | 93dc    | 1CUN       | 1CUNEBjYrCn2y1SdiUMohaKUI4wpP326Lb  |
| P 3  | 3   | 3    | 7    | e   | 5ded   | 3f69    | 19Ze       | 19ZewH8KK1PDbSndJ97FP4E1CjTRaZMZQA  |
| P 4  | 4   | 4    | 8    | e   | 9652   | 8127    | 1Ehq       | 1EhqbyUMvvs7BfL8goY6qcPbD6YKfPqb7e  |
| P 5  | 5   | 5    | 21   | e   | 8f9d   | 2be8    | 1E6N       | 1E6NuFjCi27W5zoXg8TRdcSRq84zJeBW3k  |
| P 6  | 6   | 6    | 49   | o   | f93e   | b45c    | 1Pit       | 1PitScNlyp2HcygadCh7FveTnfmPbfp8    |
| P 7  | 7   | 7    | 76   | e   | e219   | a973    | 1McV       | 1McVt1vMtCC7yn5b9wgX1833yCcLXzueeC  |
| P 8  | 8   | 8    | 224  | o   | dce7   | 1715    | 1M92       | 1M92tSqNmQLYw33fuBvjmeadirh1ysMBxK  |
| P 9  | 9   | 9    | 467  | e   | 7d0f   | aab4    | 1CQF       | 1CQFwcjw1dwhktkVwBttNLdtqL7ivBonGPV |
| P 10 | 10  | 10   | 514  | o   | d772   | c93b    | 1LeB       | 1LeBZP5QCwggXRtmVUVTrraqPUokylHqe   |
| P 11 | 11  | 11   | 155  | o   | f8c6   | 5acc    | 1PgQ       | 1PgQVlmst3Z314JrQn5TNiys8Hc38TcXJu  |
| P 12 | 12  | 12   | 683  | o   | 85a1   | f971    | 1DBa       | 1DBaumZxUkM4qMQRT2LVwyFJq5kDtSZQot  |
| P 13 | 13  | 13   | 216  | o   | f932   | 921e    | 1Pie       | 1Pie8Jkx8T6MGPz9Nvi3fsPkr2D8q3Gbc1  |
| P 14 | 14  | 14   | 544  | o   | 97f9   | f655    | 1ErZ       | 1ErZWg5cFCe4Vw5BzgfzB74VNLaxEiEkhk  |
| P 15 | 15  | 15   | 867  | e   | fe7c   | e2a2    | 1QCb       | 1QCbW9HwnwQwiQvVo5exhAnmfqKRCrsvw   |
| P 16 | 16  | 16   | 510  | e   | 7025   | e3dd    | 1BDy       | 1BDyrQ6WoF8VN3gSAS1iKzCpZfFnDviey   |
| P 17 | 17  | 17   | 823  | o   | b67c   | 2d52    | 1Hdu       | 1HduPEXZRdG26SUT5Yk83mLkPyjnZuJ7Bm  |
| P 18 | 18  | 18   | 669  | e   | ad1e   | 3f94    | 1GnN       | 1GnNTmTVLZiQqFLbAdp9DVdicEnB5GoERE  |
| P 19 | 19  | 19   | 535  | o   | ebfb   | 7dee    | 1Nwm       | 1NwmZRpHH4XSPsw6dsS3nrNWfL1yrJ4w    |
| P 20 | 20  | 20   | 317  | o   | b907   | 2868    | 1HsM       | 1HsMJxNiV7TLxmoF6uJNkydxPFDog4NQum  |
| P 21 | 21  | 21   | 764  | o   | 29a7   | 4f56    | 14oF       | 14oFNxucftsHiUMY8uctg6N487riuYXs4h  |
| P 22 | 22  | 22   | 503  | e   | 7ff4   | b86b    | 1CFZ       | 1CFZWk1QTQE3eS9qn61dQjV89KDJZzNcv   |
| P 23 | 23  | 23   | 802  | o   | d0a7   | 579e    | 1L2G       | 1L2GM8eE7mJWLdo3HZS6su1832NX2txaac  |
| P 24 | 24  | 24   | 676  | o   | 0959   | 9e2f    | 1rSn       | 1rSnXmr63jdCuegJFuiddJqWxUPV7Atuf7  |
| P 25 | 25  | 25   | 509  | o   | 2f39   | 3dbb    | 15Jh       | 15JhYXn6Mx3oF4Y7PcTAv2wVVAuCFQNiP   |
| P 26 | 26  | 26   | 862  | e   | bfeb   | 792f    | 1JVn       | 1JVnST957hGztonaWK6FougdtjxzHzRMMg  |
| P 27 | 27  | 27   | 941  | o   | 0c7a   | 5560    | 128z       | 128z5d7nN7PkCuX5qoA4Ys6pmxUYnE86k   |
| P 28 | 28  | 28   | 408  | o   | 1306   | b1da    | 12jb       | 12jbtzBb54r977TCwW361gCFoumpckRAPdy |
| P 29 | 29  | 29   | 894  | e   | 5a41   | adec    | 19EE       | 19EEC52krRUK1RkUAEmQdjTyHT7Gp1TYT   |
| P 30 | 30  | 30   | 084  | o   | d39c   | 8a08    | 1LHt       | 1LHtpnd8nU5VHEmkG2TMYNUjjLc992bps   |
| P 31 | 31  | 31   | 551  | o   | d805   | 2f1d    | 1LhE       | 1LhE6sCTuGae42Axl1LZB7L96yi9irEBE   |
| P 32 | 32  | 32   | 814  | e   | 9e42   | fdc9    | 1FRo       | 1FRoHA9xewq7DjrZ1pswJVeTer8gHRqEvR  |
| P 33 | 33  | 33   | 912  | o   | 4e15   | 3977    | 187s       | 187swfMjz1G54ycVU56B7jZFHFtNVQFDiu  |
| P 34 | 34  | 34   | 157  | o   | f6d6   | 1bfa    | 1PWA       | 1PWABE7oUahG2AFFQhvhViQovnCra4Ev7Q  |
| P 35 | 35  | 35   | 792  | e   | f6d8   | df25    | 1PWC       | 1PWCx5fovoEaoBowAvF5k91m2Xat9bMgwb  |
| P 36 | 36  | 36   | 980  | e   | 74b1   | 3d37    | 1Be2       | 1Be2UF9NLfyLFbhm3TCbmuocc9N1Kduc1i  |
| P 37 | 37  | 37   | 595  | e   | 28c3   | 021d    | 14ix       | 14ixhn8b6GajVwegZHJ18vJLHhntcpL4dex |
| P 38 | 38  | 38   | 592  | o   | b190   | 9364    | 1HBT       | 1HBTApAFA9B2YzW3G2YKSMctb3dVnjuNe2  |
| P 39 | 39  | 39   | 937  | e   | 0b30   | 0876    | 122A       | 122AJhKLEfkFBaGAd84pLp1kfE7xK3GdTv  |
| P 40 | 40  | 40   | 950  | o   | 95a1   | a82a    | 1EeA       | 1EeAxcprB2PpCnr34VfZdFrkUwuxyiNEFv  |
| P 41 | 41  | 41   | 147  | o   | d156   | 2329    | 1L5s       | 1L5sU9qvJeuwQUdt4y1eiLmqFxFkjtHr3E  |
| P 42 | 42  | 42   | 463  | o   | 8efb   | ae23    | 1E32       | 1E32GPWgdYeyQac4aJxm9HVoLrrEYPnM4N  |
| P 43 | 43  | 43   | 825  | e   | f920   | 86f7    | 1PiF       | 1PiFuqGpG8yGM5v6rNHWs3TjSg6awgEGA1  |
| P 44 | 44  | 44   | 071  | e   | 80df   | 791e    | 1CkR       | 1CkR2uS7LmFwc3T2jv8C1Bhwb5mQaoxedF  |
| P 45 | 45  | 45   | 597  | e   | f022   | 0753    | 1Nti       | 1NtiLNGegHWE3Mp9g2JpKgx6wUg4TW7bbk  |
| P 46 | 46  | 46   | 612  | o   | 9a01   | 5dac    | 1F3J       | 1F3JRMWudBaj48EhwcHDDpeuy2jwaACNxjP |
| P 47 | 47  | 47   | 170  | e   | f828   | 07d4    | 1Pd8       | 1Pd8Vvt49sHkSmqrQip61RsvwmXCZ6ay7Z  |
| P 48 | 48  | 48   | 443  | e   | 8661   | a2b2    | 1DFY       | 1DFYhaB2J9q1LLZJWKTnscPwos9VBqDHzv  |
| P 49 | 49  | 49   | 525  | e   | 0d2f   | fdf3    | 12Ci       | 12CiUhyVTH33w3SPUBgcpMoqnAaV4wCf    |
| P 50 | 50  | 50   | 764  | o   | de08   | 6a47    | 1MEz       | 1MEzite4ReNuWaL5Ds17ePKt2dCxEofwk   |
| P 51 | 51  | 51   | 876  | e   | ef64   | dbe7    | 1Npn       | 1NpnQyZ7x24ud82b7wiRNvPm6N8bqGQnaS  |
| P 52 | 52  | 52   | 700  | o   | 36af   | 3ae7    | 15z9       | 15z9c9sVpu6fwNiK7dMAFgMYSK4GqsGZim  |
| P 53 | 53  | 53   | 124  | e   | 2f48   | 9551    | 15K1       | 15K1YKJMiJ4fpestVUCByoz334rHmknxmT  |
| P 54 | 54  | 54   | 707  | o   | cb66   | a092    | 1KYU       | 1KYUv7nSvXx4642TkeuC2SndTk326uUpFy  |
| P 55 | 55  | 55   | 460  | o   | db53   | 2fa2    | 1Lzh       | 1LzhS3k3e9Ub8i2W1V8xQfDB8n2MYCHPCA  |
| P 56 | 56  | 56   | 575  | o   | 4821   | cccc    | 17aP       | 17aPYR1m6pVAacXg1PTDDU7XafvK1dxvhi  |
| P 57 | 57  | 57   | 492  | e   | 3286   | c2a1    | 15c9       | 15c9mPGLku1HuW9LrtBf4jchVPBUT8txKz  |
| P 58 | 58  | 58   | 049  | o   | 8c2a   | a60f    | 1Dn8       | 1Dn8NF8qDyyfHmktmuoQLGyjWmZxgvosXf  |
| P 59 | 59  | 59   | 191  | e   | b14e   | 0210    | 1HAX       | 1HAX2n9Uruu9YDt4cqRgYcvTgVzj1rbUyt  |
| P 60 | 60  | 60   | 382  | o   | cdff   | b9c5    | 1Kn5       | 1Kn5h2qpgw9mWE5jKpk8PP4qvJ1QVv8su   |
| P 61 | 61  | 61   | 982  | e   | 6813   | 0cbd    | 1AVJ       | 1AVJKws9AskraJLGHAPiaZCprrD1U6AB    |
| P 62 | 62  | 62   | 062  | o   | e266   | b779    | 1Me6       | 1Me6EfpwZK5kQziBwBfVLiHjaPGxCKLoJi  |
| P 63 | 63  | 63   | 768  | o   | ef58   | c50e    | 1NpY       | 1NpYjtLira16LfGbGwZJ5JbDPH3ai9bjf4  |
| P 64 | 64  | 64   | 628  | o   | 3ee4   | 48a4    | 16jY       | 16jY7qLJnxb7CHZyqBP8qca9d51gAjyXQN  |
| P 65 | 65  | 65   | 855  | e   | 52e7   | 0137    | 18ZM       | 18ZMbwUFLMhoZBbfpCjUJQTCMCbktshgpe  |
| P 66 | 66  | 66   | 726  | e   | 20d4   | b8a5    | 13zb       | 13zb1hQbWvsc2S7ZTznP2G4undNNpdh5so  |
| P 67 | 67  | 67   | 302  | e   | 7394   | 9371    | 1BY8       | 1BY8GQbnueYofwSuFAT3USAHgjPrkxDdw9  |
| P 68 | 68  | 68   | 825  | o   | e0b8   | 8cfc    | 1MVD       | 1MVDYgVaSN6iKKEsbzRUAYFrYJadLYzvz   |
| P 69 | 69  | 69   | 804  | e   | 61eb   | d5aa    | 19vk       | 19vkiEaJfhuZ8bs8Zu2jgmC6oqZbwqhXhG  |
| P 70 | 70  | 70   | 481  | e   | 5db8   | 1b10    | 19YZ       | 19YZECXj3SxEZMoUeJ1y1Psw8xANe7M7QR  |
| P 75 | 75  | 75   | 367  | o   | badf   | e5cf    | 1J36       | 1J36UjUByGroXcCvmj13U6uwaVv9caEeAt  |
| P 80 | 80  | 80   | 456  | o   | 6fe5   | 6fab    | 1BCF       | 1BCF6rHUW6m3iH2ptsvnjgLRua1PQqPeLe  |
| P 85 | 85  | 85   | 920  | o   | cd03   | c281    | 1Kh2       | 1Kh22PvXERd2xpTqK3ur6pPEqFeckCJfAr  |
| P 90 | 90  | 90   | 863  | o   | d06b   | 674b    | 1L12       | 1L12FHH2FHjvTviyanuiFVfmzCy46RRATU  |
| P 95 | 95  | 95   | 212  | e   | 5ed8   | 6d88    | 19eV       | 19eVSDuizyDxxhohGh8K19WY9KsHdSwoQC  |
| P100 | 100 | 100  | 142  | o   | c7a7   | bc6e    | 1KCg       | 1KCgMv8fo2TPBpdDVi9jgmMmcne9uSNJ5F  |

|      |     |     |   |      |      |      |                                    |
|------|-----|-----|---|------|------|------|------------------------------------|
| P105 | 105 | 435 | o | 7c95 | 50b0 | 1CMj | 1CMjscKB3QW7SDyQ4c3C3DEUHiHRhiZVib |
| P110 | 110 | 748 | o | 0e5f | f207 | 12Jz | 12JzYkKN76xkwvcPT6AWKZtGX6w2LAgsJg |
| P115 | 115 | 950 | e | ea0f | 39a4 | 1NLb | 1NLbHuJebVwUZ1XqDjsAyfTRUPwDQbemfv |
| P120 | 120 | 323 | e | 4b46 | 308e | 17s2 | 17s2b9Ksz5y7abUm92cHwG8jEPCzK3dLnT |
| P125 | 125 | 420 | e | f707 | 5fc8 | 1PXA | 1PXAYUB8ZoH3WD8n5zoAthYjN15yN5CVq5 |
| P130 | 130 | 577 | o | a249 | cf75 | 1Fo6 | 1Fo65aKq8s8iquMt6weF1rku1mowVED5Ua |

=== step-by-step (P1..P20) ===

```
--- P1 ---
d (dec)      = 1
d (hex)      = 1
d*G          px = 5506626302227734366957871889516853432625060345377594175500187360389116729240
              py = 32670510020758816978083085130507043184471273380659243275938904335757337482424 (even)
compressed   = 0279be667ef9dcbbac55a06295ce870b07029bfcd2dce28d959f2815b16f81798
SHA256       = 0f715baf5d4c2ed329785cef29e562f73488c8a2bb9dbbc5700b361d54b9b0554
hash160      = 751e76e8199196d454941c45d1b3a323f1433bd6
checksum     = 510d1634
address      = 1BgGZ9tcN4rm9KBzDn7KprQz87SZ26SAMH
53125 match px=True py=True
```

```
--- P2 ---
d (dec)      = 3
d (hex)      = 3
d*G          px = 112711660439710606056748659173929673102114977341539408544630613555209775888121
              py = 25583027980570883691656905877401976406448868254816295069919888960541586679410 (even)
compressed   = 02f9308a019258c31049344f85f89d5229b531c845836f99b08601f113bce036f9
SHA256       = eae10cdd2f289bdad44615809cb422d2fabe9622ed706ad5d9d3ffd2cdd1c001
hash160      = 7dd65592d0ab2fe0d0257d571abf032cd9db93dc
checksum     = ac074bec
address      = 1CUNEBjYrCn2y1SdiUMohaKUi4wpP326Lb
53125 match px=True py=True
```

```
--- P3 ---
d (dec)      = 7
d (hex)      = 7
d*G          px = 41948375291644419605210209193538855353224492619856392092318293986323063962044
              py = 48361766907851246668144012348516735800090617714386977531302791340517493990618 (even)
compressed   = 025cddf0646e5db4eaa398f365f2ea7a0e3d419b7e0330e39ce92bddedcac4f9bc
SHA256       = a2039429ca2c2f2bcc0725a1682aeceb3ac1b8e77248c34fa57fcdef29d01c53
hash160      = 5dedfbf9ea599dd4e3ca6a80b333c472fd0b3f69
checksum     = 40910b7f
address      = 19ZewH8Kk1PDbSndJ97FP4EiCjTRaZMZQA
53125 match px=True py=True
```

```
--- P4 ---
d (dec)      = 8
d (hex)      = 8
d*G          px = 2126057306151627953595685090280431278183829487175876377991189246716355947009
              py = 41749993296225487051377864631615517161996906063147759678534462689479575333124 (even)
compressed   = 022f01e5e15cca351daff3843fb70f3c2f0a1bdd05e5af888a67784ef3e10a2a01
SHA256       = be2b01947193835b2a70e0bed841b4dd8926e75f6a7427ba3d90a1774beacac6
hash160      = 9652d86bedf43ad264362e6e6ba6eb764508127
checksum     = 4e02d7e9
address      = 1EhqbyUMvvs7BfL8goY6qcPbd6YKfPqb7e
53125 match px=True py=True
```

```
--- P5 ---
d (dec)      = 21
d (hex)      = 15
d*G          px = 24049875635381557237058143631624836741422505207761609709712554171343558302165
              py = 22669890352939653242079781319904043788036611953081321775127194249638113810828 (even)
compressed   = 02352bbf4a4cdd12564f93fa332ce333301d9ad40271f8107181340aef25be59d5
SHA256       = 8e46c2c3a5c927c9d23d5aa249c838a91256174db96c61c9e5d340aae9962e6
hash160      = 8f9dff39a81ee4abcbad2ad8bafff090415a2be8
checksum     = c13898f3
address      = 1E6NuFjCi27W5zoXg8TRdcSRq84zJeBW3k
53125 match px=True py=True
```

```
--- P6 ---
d (dec)      = 49
d (hex)      = 31
d*G          px = 109846273198995465046214558810696509369395695668678477567979041624391316796720
              py = 101711840700253487202274292900077312932162220876743713596510869779955653950519 (odd)
compressed   = 03f2dac991cc4ce4b9ea44887e5c7c0bce58c80074ab9d4dbaeb28531b7739f530
SHA256       = 5d80f01b950b9ec98536759b4b28ceadf4a4fe915642d26d0a0f42efdc16c890
hash160      = f93ec34e9e34a8f8ff7d600cdad83047b1bcb45c
checksum     = 7c419e15
address      = 1PitScNLyp2HCygzadCh7FveTnfmPbfp8
53125 match px=True py=True
```

```
--- P7 ---
d (dec)      = 76
d (hex)      = 4c
d*G          px = 67990777350807742601404023085488753338978132210207570380931332031538910989091
              py = 8579252808009126161701413187313528408286619337671538805045464108358228475888 (even)
compressed   = 0296516a8f65774275278d0d7420a88df0ac44bd64c7bae07c3fe397c5b3300b23
SHA256       = 2d30f8a53167344dca3344e2f526ecfe952b22ec53f485fcb79c32be8ff7d12
hash160      = e2192e8a7dd8dd1c88321959b477968b941aa973
checksum     = 4f595951
address      = 1McVt1vMtCC7yn5b9wgX1833yCcLXzueeC
```

```
53125 match px=True py=True

--- P8 ---
d (dec)      = 224
d (hex)      = e0
d*G          px = 3951620832557057103848397037937330228967155552456836977045923180329637950740
              py = 95473263819178136898129622939316779429655166328021079992747693728567241822551  (odd)
compressed   = 0308bc89c2f919ed15885c35600844d49890905c79b357322609c45706ce6b514
SHA256       = 557f3f82c4ee71eb63b44b4c982c5284b28c48ec720dbce50262f780b9065190
hash160      = dce76b2613052ea012204404a97b3c25eac31715
checksum     = 5d79aa30
address      = 1M92tSqNmQLYw33fuBvjmeadirh1ysMBxK
53125 match px=True py=True

--- P9 ---
d (dec)      = 467
d (hex)      = 1d3
d*G          px = 30474780960141219921807662305060768687772753653098971817340717083315339616503
              py = 61319752531495161980867220483083814171916864550579329528527835522023988931626  (even)
compressed   = 0243601d61c836387485e9514ab5c8924dd2cfd466af34ac95002727e1659d60f7
SHA256       = bd6e9f3debdb4aae95e11c2b883fe21ace4b0860ea5676b5e05072405dae18ae
hash160      = 7d0f6c64afb419bbd7e971e943d7404b0e0daab4
checksum     = 89291a1c
address      = 1CQFwcjw1dwhktVWBttNLdtQL7ivBonGPV
53125 match px=True py=True

--- P10 ---
d (dec)      = 514
d (hex)      = 202
d*G          px = 75827354541827746090915769557428734838962266972502112805530189196736906960333
              py = 96570036462163023759491147881411408314153451204876534252590428078766327205003  (odd)
compressed   = 03a7a4c30291ac1db24b4ab00c442aa832f7794b5a0959bec6e8d7fee802289dcd
SHA256       = 85df64d76f2f7e5a90e36ec0b0b6da5e8621c3773e0ba594bb9fa75068e22bb
hash160      = d7729816650e581d7462d52ad6f732da0e2ec93b
checksum     = abcb389d
address      = 1LeBZP5QCwwgXRtmVUVTVrraqPUokyLHqe
53125 match px=True py=True

--- P11 ---
d (dec)      = 1155
d (hex)      = 483
d*G          px = 62881537490095887219700399231850374465934748632261627566823792868512270937739
              py = 13147057280235047456932561657169493651618639731944681771959256397716727593785  (odd)
compressed   = 038b05b0603abd75b0c57489e451f811e1afe54a8715045cdf4888333f3ebc6e8b
SHA256       = 41ced0f93ba060a4266f2c955d42484aa3e4f169e9b26ab35aa20d798b6e6d59
hash160      = f8c698da3164ef8fa4258692d118cc9a902c5acc
checksum     = 6078c6be
address      = 1PgQVLmst3Z314JrQn5TNiys8Hc38TcXJu
53125 match px=True py=True

--- P12 ---
d (dec)      = 2683
d (hex)      = a7b
d*G          px = 62873230362895154212630786022242768049765907075202803363316800197224716728348
              py = 44287605905639058437764882210076840571916049925678373415089141948008474303433  (odd)
compressed   = 038b00fcbfc1a203f44bf123fc7f4c91c10a85c8eae9187f9d22242b4600ce781c
SHA256       = d4e3872b0dc62b65c97616f493f899b4ba79422428e6bdbaf150ecd53d217
hash160      = 85a1f9ba4da24c24e582d9b891dacbd1b043f971
checksum     = 881319cb
address      = 1DBaumZxUkM4MQRt2LVWfJq5kDtSZQot
53125 match px=True py=True

--- P13 ---
d (dec)      = 5216
d (hex)      = 1460
d*G          px = 77279535011188034078516592160190105278519663645387310740933932046966334391965
              py = 21187662028949108981404481381270595331275020854904259675645205037069861834225  (odd)
compressed   = 03aadaaab1db8d5d450b511789c37e7cfef0eb8b3e61a57a34166c5edc9a4b869d
SHA256       = e7f16e16631ec0261aa102d0e43894e7cadd05bd80edcf0aaf6d18073627a27a
hash160      = f932d0188616c964416b91fb9cf76ba9790a921e
checksum     = 4f507c8e
address      = 1Pie8JkxBT6MGPz9Nvi3fsPkr2D8q3GBc1
53125 match px=True py=True

--- P14 ---
d (dec)      = 10544
d (hex)      = 2930
d*G          px = 81843657467215387495831064399406566332608612828066495589872347702422424340313
              py = 54485907183099442435241830545487744604431707282385107218140713089194934994051  (odd)
compressed   = 03b4f1de58b8b41afe9fd4e5ffbdafaeab86c5db4769c15d6e6011ae7351e54759
SHA256       = 7334d1e5de4a185ad636f2d835c514aa7dec265c3b047171a1a01a8449d4bd2e5
hash160      = 97f9281a138379d72ac52a6a3e9e8b9a4a4f655
checksum     = 468715ff
address      = 1ErZwg5cFCe4Vw5BzgffzB74VNLaxEiEkKh
53125 match px=True py=True

--- P15 ---
d (dec)      = 26867
d (hex)      = 68f3
d*G          px = 115179987075229906903259602194857104773352008368264866291918996865985783521068
              py = 90040672269639781867601170828097460642155183514693739441068299505511569496110  (even)
```

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compressed = 02fea58ffcf49566f6e9e9350cf5bca2861312f422966e8db16094beb14dc3df2c
SHA256      = e0ff68c71230fb46a9d4deafb0cdaae47c8194ba9274a15d7ce8c89bfaca48ba
hash160     = fe7c45126731f7384640b0b0045fd40bac72e2a2
checksum    = 9849ab37
address     = 1QCbw9HwnwQwiQqVo5exhAnmfqKRrCRsvW
53125 match px=True py=True

--- P16 ---
d (dec)     = 51510
d (hex)     = c936
d*G         px = 71261119111627271268814301300816751730184609798722377831958734823956300761978
              py = 59596021694081902311438976720015207351957060174440251698640379092733344351048  (even)
compressed  = 029d8c5d35231d75eb87fd2c5f05f65281ed9573dc41853288c62ee94eb2590b7a
SHA256      = e60cf41a9a515452a1f757374158d0e1a97d3764ae048604ce892d7b08747693
hash160     = 7025b4efb3ff42eb4d6d71fab6b53b4f4967e3dd
checksum    = 24d10b05
address     = 1BDyrQ6WoF8VN3g9SAS1iKZcPzFfnDVieY
53125 match px=True py=True

--- P17 ---
d (dec)     = 95823
d (hex)     = 1764f
d*G         px = 28680425603065441115290152280488699295594719576682295038107311098738046478531
              py = 25891784853872784916698406928566284655180782352356802345945352320731968056623  (odd)
compressed  = 033f688bae8321b8e02b7e6c0a55c2515fb25ab97d85fda842449f7bfa04e128c3
SHA256      = fb3d988e5a8e695f81c148bec60e05b237e9dcce95ea963ef068537e1207ad41
hash160     = b67cb6edeabc0c8b927c9ea327628e7aa63e2d52
checksum    = f4a7f370
address     = 1HduPEXZRdG26SUT5Yk83mLkPyjnZuJ7Bm
53125 match px=True py=True

--- P18 ---
d (dec)     = 198669
d (hex)     = 3080d
d*G         px = 5831721406635063654683117253506846208566679410118154188144422804291796683505
              py = 61770026510604113021686089777374557608174938587526483912023179154647723408296  (even)
compressed  = 020ce4a3291b1d72e1a7bf73ee87d30a6bdbc72b20771e7dffff40d0db755cd4af1
SHA256      = 19b9cafbb1cababfd84681777cdcb38f47a275698dab5aa9bf8904a01b2aed464
hash160     = ad1e852b08eba53df306ec9daa8c643426953f94
checksum    = d1c893f1
address     = 1GnNTmTVLZiqQfLbAdp9DVdicEnB5GoERE
53125 match px=True py=True

--- P19 ---
d (dec)     = 357535
d (hex)     = 5749f
d*G         px = 60338245119404298188752055094151291956247541820833238489095541627141964729881
              py = 31432635976074881250581475377260990518299484384225785790788933914681598133873  (odd)
compressed  = 0385663c8b2f90659e1ccab201694f4f8ec24b3749cfe5030c7c3646a709408e19
SHA256      = 39b5324342c3490e0cbafd32fb7560bd3bd5cc74ead2c7a19d35a5e165a35e9e
hash160     = ebfbe6819fcdebab061732ce91df7d586a037dee
checksum    = 7fbb6c04
address     = 1NwmZRPHH4XSPsw6dsS3nrNwFL1yrJJ4w
53125 match px=True py=True

--- P20 ---
d (dec)     = 863317
d (hex)     = d2c55
d*G         px = 27269999313721489064003814688126459473916071371077568700989584042678664122236
              py = 61518159794260916008520957000838288631389138584881336375456078921081169337009  (odd)
compressed  = 033c4a45cbbd643ff97d77f41ea37e843648d50fd894b864b0d52f6bc62f6454f7c
SHA256      = 7dc838f915b684beac241be8ed919e71a7a8e1d15befda4df3b9154f19ba4fb9
hash160     = b907c3a2a3b27789dfb509b730dd47703c272868
checksum    = 2e201588
address     = 1HsMJxNiV7TLxmoF6uJNkydxPFDog4NQuM
53125 match px=True py=True

=== hash160 hex – vertical nibble stack (40 nibbles = 20 bytes) ===
h160[-39] P 1-P40 77598fed7df8f9f7baeb27d02b015dd94ff72b09
h160[-39] P 41-P120 d8f8f9f80de32cd438bc6ee3527e65b6cd5c70e4
h160[-39] P125-P130 fa
h160[-38] P 1-P40 5dd6f92cd78597e06db99f09ffc3a38ee66481b5
h160[-38] P 41-P120 1e900a86def6fbb82c1d82fe20301dafd0e7ceab
h160[-38] P125-P130 72
h160[-37] P 1-P40 1de5931e07ca3f7271f0afa53e7049041ddbc93a
h160[-37] P 41-P120 5f2d2026206a4652824f165eed9bebbe06da9504
h160[-37] P125-P130 04
h160[-36] P 1-P40 e6d2de97f26129c5ceb774799ba61c5256813001
h160[-36] P 41-P120 6b0f2181f84f86316ae836847448b8f53b875ff6
h160[-36] P125-P130 79
h160[-35] P 1-P40 75fdfc26699fd24bb8ec859e6bab64f6e7ce0e45
h160[-35] P 41-P120 2845520c511677d466de34a1653a8c8ac62b732e
h160[-35] P125-P130 92
h160[-34] P 1-P40 65b8f3ebc8890854656323d8b7f9c7605de0f2f6
h160[-34] P 41-P120 e544b20b3b95069c0035e6f33a72adb31e22dcb1
h160[-34] P125-P130 22
h160[-33] P 1-P40 e9f6348261db111ee28a10f0236ec0f11721bd2c
h160[-33] P 41-P120 bfcef65537c9e3b5e71c1db3a6bb5a06e213b470
h160[-33] P125-P130 58
h160[-32] P 1-P40 829b9ea646aa8a2fdb12331195c4942e892294ad
h160[-32] P 41-P120 3971c0d696fecb9f1479b6d7aba05de602f605a

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h160[-32] P125-P130 65  
h160[-31] P 1-P40 1deea971a634816be09ac782b6af165e985b1072  
h160[-31] P 41-P120 7c9f6d4d6fffd5fd64f65b899d73ec33e2656f675  
h160[-31] P125-P130 a2  
h160[-30] P 1-P40 90ad8ed3f51d6373a8f3a79122af461a73fe1c91  
h160[-30] P 41-P120 3526801968db471938f02479d6de8a4f663bd364  
h160[-30] P125-P130 a0  
h160[-29] P 1-P40 9a5f13d0b06a183fbecba4ff7d7584fe5bf18fab  
h160[-29] P 41-P120 5b41a1bdc47efffa953dbb1c2d16620896dd9b1  
h160[-29] P125-P130 05  
h160[-28] P 1-P40 1b94e4854e42631fcbd29ee334e6fe7d2fb5ed04  
h160[-28] P 41-P120 75e26c0f60f9ad3e6cba0f0f156bba86c15907da  
h160[-28] P125-P130 21  
h160[-27] P 1-P40 9293ead215e4c8f40ae7ef162555414cd7d2de2a  
h160[-27] P 41-P120 fd5fe5f05ea44ea96929b695a3d70098e228740e  
h160[-27] P125-P130 7a  
h160[-26] P 1-P40 6fda48de98fc9772c5b7e7aa4441ad7210e11e76  
h160[-26] P 41-P120 9b52114a74d4b68fa0cd9042a51750e4997b349e  
h160[-26] P125-P130 d9  
h160[-25] P 1-P40 deddaf1ab182693e83a8aade31233e94e4cede29  
h160[-25] P 41-P120 e22f71f686750534bc92024f479f826ab548328c  
h160[-25] P125-P130 c0  
h160[-24] P 1-P40 4042b8c0bdf44d8bbdb9855a0b4a7be4a5e5a27d  
h160[-24] P 41-P120 6d5ce3e1e20349b3e4b23f2df08c4d7f8e2ab996  
h160[-24] P125-P130 40  
h160[-23] P 1-P40 5de6cf81d7ae47449f0d22c103b477bef0cd7c0e  
h160[-23] P 41-P120 f5c58dd1f2f880f2dd6433fa037bb909c6ab92b  
h160[-23] P125-P130 32  
h160[-22] P 1-P40 4034bf827445126d236f4f33d267b6d14f182e29  
h160[-22] P 41-P120 c56b7f401a76c6976ae4e2c68ce0b1ebe32f861e  
h160[-22] P125-P130 7e  
h160[-21] P 1-P40 92c3a732e6286a46701baf0b0a366c85427dee76  
h160[-21] P 41-P120 491dc6c1129319455b12dbea1e6370c9341538f3  
h160[-21] P125-P130 cb  
h160[-20] P 1-P40 45a6dd209252bc0dc675cf63c0d8795f49077069  
h160[-20] P 41-P120 172cd6831c943ae0357ef39219642361c592b2cf  
h160[-20] P125-P130 bf  
h160[-19] P 1-P40 176226147d8d95b79e30fdcb52342331d5f5c75e  
h160[-19] P 41-P120 c3008cd29d446ee7fa0394055e65765f3717c5ea  
h160[-19] P125-P130 b4  
h160[-18] P 1-P40 cdaea09415691201ec2908ca9b11c3b7fc8e423b  
h160[-18] P 41-P120 ba75b8c87f3e57c0ab3e8b7c7025d5c3e8db6f9c  
h160[-18] P125-P130 5c  
h160[-17] P 1-P40 4586d050e29bfabfa9cb807dba7b8133d927f936  
h160[-17] P 41-P120 209c5aab8359dfac12dc35298b9fd96be22eddb7  
h160[-17] P125-P130 38  
h160[-16] P 1-P40 570e8c949a28b60a3de7010316faaed76c9485f7  
h160[-16] P 41-P120 3479e835c642407e79ef0a18c2c366c6adbb63f0  
h160[-16] P125-P130 96  
h160[-15] P 1-P40 d1b6bdba4dd99a0b2a932146760ce605bb8594e1  
h160[-15] P 41-P120 512d64b5a0e026da85aca6f34159576cb777d9e9  
h160[-15] P125-P130 f4  
h160[-14] P 1-P40 1a3eaa493611c3467a102991c4f753faf8caca16  
h160[-14] P 41-P120 92c2030a61bd58d14de822b496fdbd1f8e7275bc  
h160[-14] P125-P130 9a  
h160[-13] P 1-P40 bb3bfd77df1dfe5b68dda87c3d8bd74df23202e8  
h160[-13] P 41-P120 f821a86f9781989e57970cbe1e05e7dcdae445e2  
h160[-13] P125-P130 55  
h160[-12] P 1-P40 3f3af87b778a79f52cfdb1b1a1fa35bb38f61bd6  
h160[-12] P 41-P120 4d5aec0783e4447227e80a10c381df39008d7435  
h160[-12] P125-P130 55  
h160[-11] P 1-P40 a0c6f39343cc6ed386749035b41426909dcc6e44  
h160[-11] P 41-P120 dc38391efdf3cd0ca2f1c39711c488a7f68d7376  
h160[-11] P125-P130 4a  
h160[-10] P 1-P40 334e006c02cbb84be4d7d35dc0f884be9a2e48ef  
h160[-10] P 41-P120 37c3bf158fafc9bb5937fca4b977d59218da15ff  
h160[-10] P125-P130 7c  
h160[-9] P 1-P40 227b94824d9dab047357f48a1b9d7c22fa8d79f4  
h160[-9] P 41-P120 e59ae706aae17ad4dd36da35c9072d7db649116b  
h160[-9] P125-P130 2b  
h160[-8] P 1-P40 3c2707b5baa199bfa480cb4c85f6a26ceba85efc  
h160[-8] P 41-P120 d1fa1a7adf27d081f1a7fe5a74e4414669342437  
h160[-8] P125-P130 47  
h160[-7] P 1-P40 fdf64b9e009b7aa4a26382123aa9d55db2e46738  
h160[-7] P 41-P120 a0c88123d4273f5dcbbf2319cf3550938e666f8d  
h160[-7] P125-P130 15  
h160[-6] P 1-P40 19d4114aee0094c966ac35565545a7600665eb0b  
h160[-6] P 41-P120 409dfecf7a38d8b555878fe7d3210e5c89d03d3a  
h160[-6] P125-P130 6b  
h160[-5] P 1-P40 4d055b1c02240a76390248d1b701add8458e8a98  
h160[-5] P 41-P120 07f213Fef9c6274b9418c9f4557fd134cddb0be6  
h160[-5] P125-P130 7b  
h160[-4] P 1-P40 3bb0aca3dec3a427e5371ca8d4d6f69e4f6aa312  
h160[-4] P 41-P120 35002db7f725699d4e1791727d6c67bfa851a669  
h160[-4] P125-P130 24  
h160[-3] P 1-P40 39382ba1ac5f9fee23724b59375ba82f31d3090a  
h160[-3] P 41-P120 2a87050af6d39a2cca0b0bc40b98d1e6c66b5f33  
h160[-3] P125-P130 5c  
h160[-2] P 1-P40 b3f1b497a9a92623dfd8f87ed951dafd9b9fd2388  
h160[-2] P 41-P120 3e697d72daba50fc2629c758183c5b5f27dc0290  
h160[-2] P125-P130 ff

